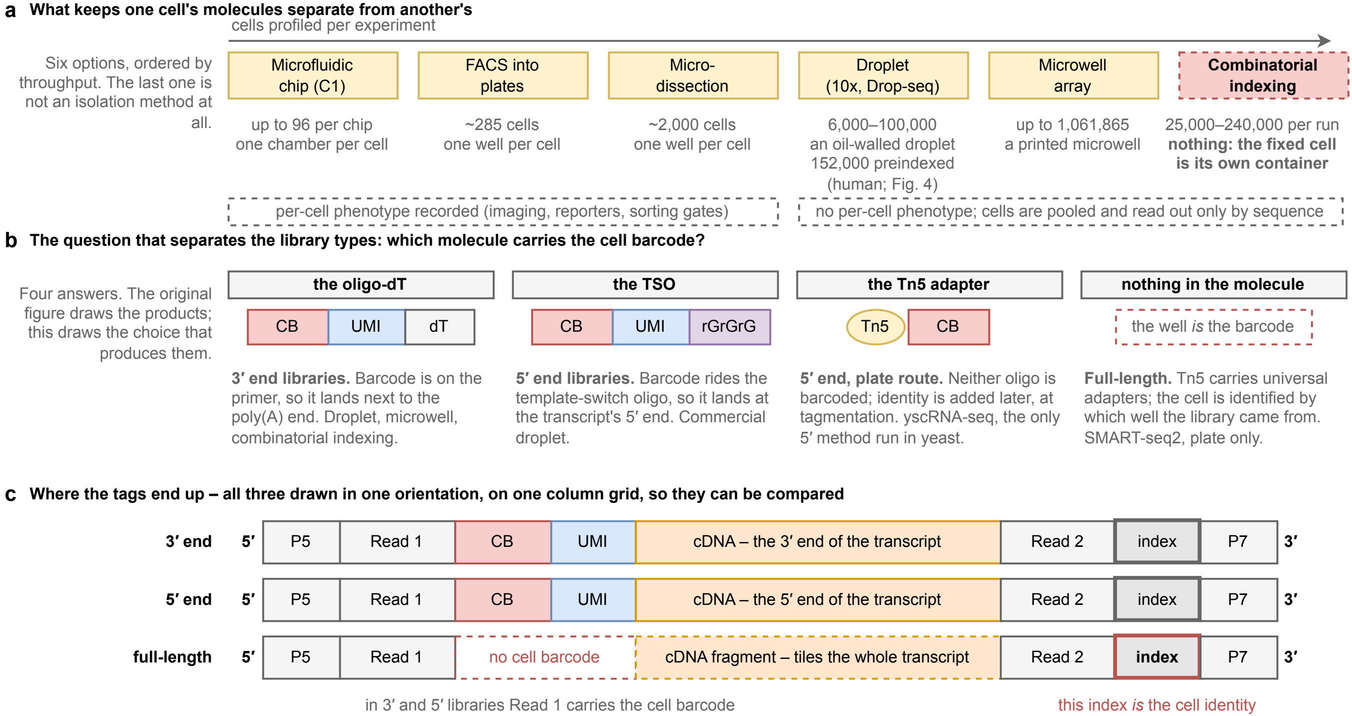
Yeast scRNA-seq methods: what separates the cells, and which molecule carries the cell barcode

A redrawing of Fig. 1 of Nadal-Ribelles 2024, reorganized around the one question that separates the library types. Every molecule is drawn in a single fixed orientation, 5′ left to 3′ right with P5 always on the left, and colors are those of Figs. 1 and 2.



Read panel b and panel c together and the taxonomy collapses to one rule: a cell barcode is only needed where cells are pooled before sequencing. Droplet, microwell and combinatorial methods pool early, so identity has to be written into the molecule, and it goes wherever the barcoded oligo sits – the poly-dT end for 3′ libraries, the template-switch end for 5′. Plate methods never pool until the very end, so the molecule needs no barcode at all and the Illumina sample index does the work, one index per well. That is the same accounting split-pool uses when its sublibrary index becomes barcode round 4.

Segments transcript / cDNA | cell barcode (CB) | UMI | TSO rGrGrG | container / enzyme | handle / adapter | Illumina sample index | red dashed = absent, or doing an unexpected job